

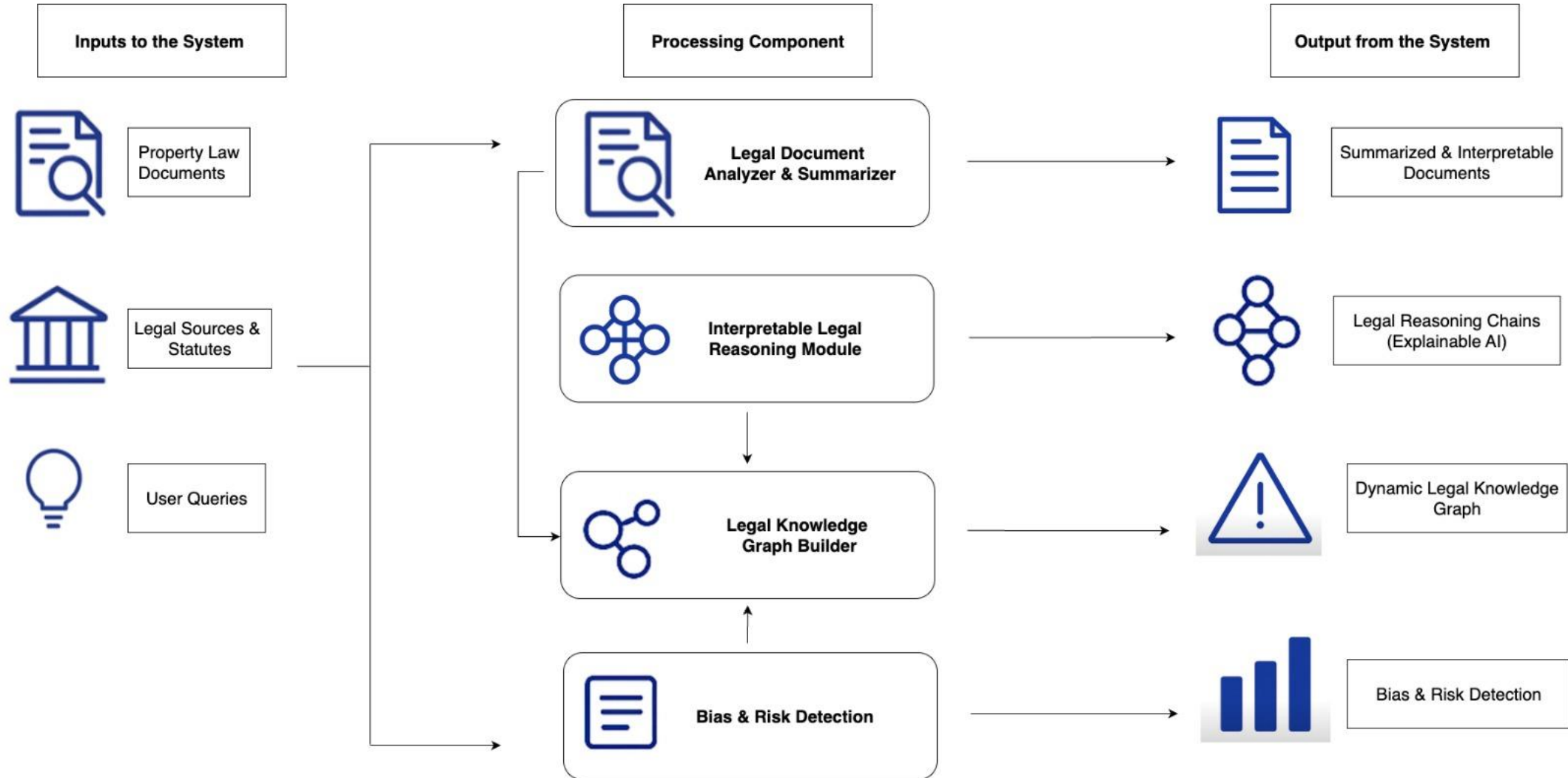
A Knowledge-Driven AI Framework for Legal Understanding and Trust

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A Knowledge-Driven AI Framework for Legal Understanding and Trust



A Knowledge-Driven AI Framework for Legal Understanding and Trust





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Information Technology Specialized in Data Science

Visual-Integrated Multi Perspective Legal Summarizer

Research Gap

- Lack multi-perspective summaries
- Plain-language outputs
- Traceable summaries
- Accurate visualizations

Solution

- multi-reference summary datasets
- legal jargon normalization,
- linking back to original clause,
- traceable visualizations (infographics, Entity relationship diagrams)

Existing Systems

- AI Pazz → Focuses on case law documents, data retrieval, contextual insights
- LankaLAW → retrieves Acts and laws in a conversational way
- **All existing systems target legal professionals and students – not normal people who need plain-language legal understanding.**

Visual-Integrated Multi Perspective Legal Summarizer

Research problem

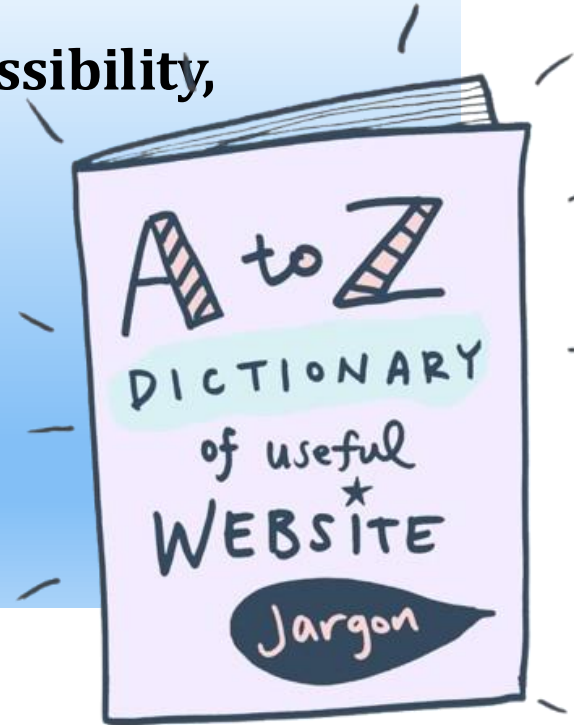
How can summarization and visualization techniques be integrated to improve non-experts' understanding of complex legal documents?

Specific Objectives

- Develop a summarization & visualization system that improves **accessibility**, **interpretability & trust**.

Sub Objectives

- Simplify legal jargon → plain language, Generate **multi-perspective summaries** (case, financial, procedural, rights/duties).
- Ensure **traceability** back to original clauses.
- Provide **visualizations** (diagrams, infographics, timelines).



Visual-Integrated Multi Perspective Legal Summarizer

Key Domain

Knowledge Utilized to Fabricate the Solution

Latest Technology

1

**Data Science /
NLP**

Legal text preprocessing, clause
segmentation, Named Entity
Recognition (NER)

spaCy, NLTK

2

**Machine
Learning / Deep
Learning**

Transformer-based summarization
for multi-perspective outputs

LegalBERT, BART,
T5, Pegasus

3

Visualization

Graph-based representation,
infographics, interactive
visualization of legal clause
relationships

Graphviz, Plotly,
D3.js

Visual-Integrated Multi Perspective Legal Summarizer

Evaluation

Quantitative:

- ROUGE scores for summarization quality
- Precision–Recall for key insights
- Response time measurements

Qualitative:

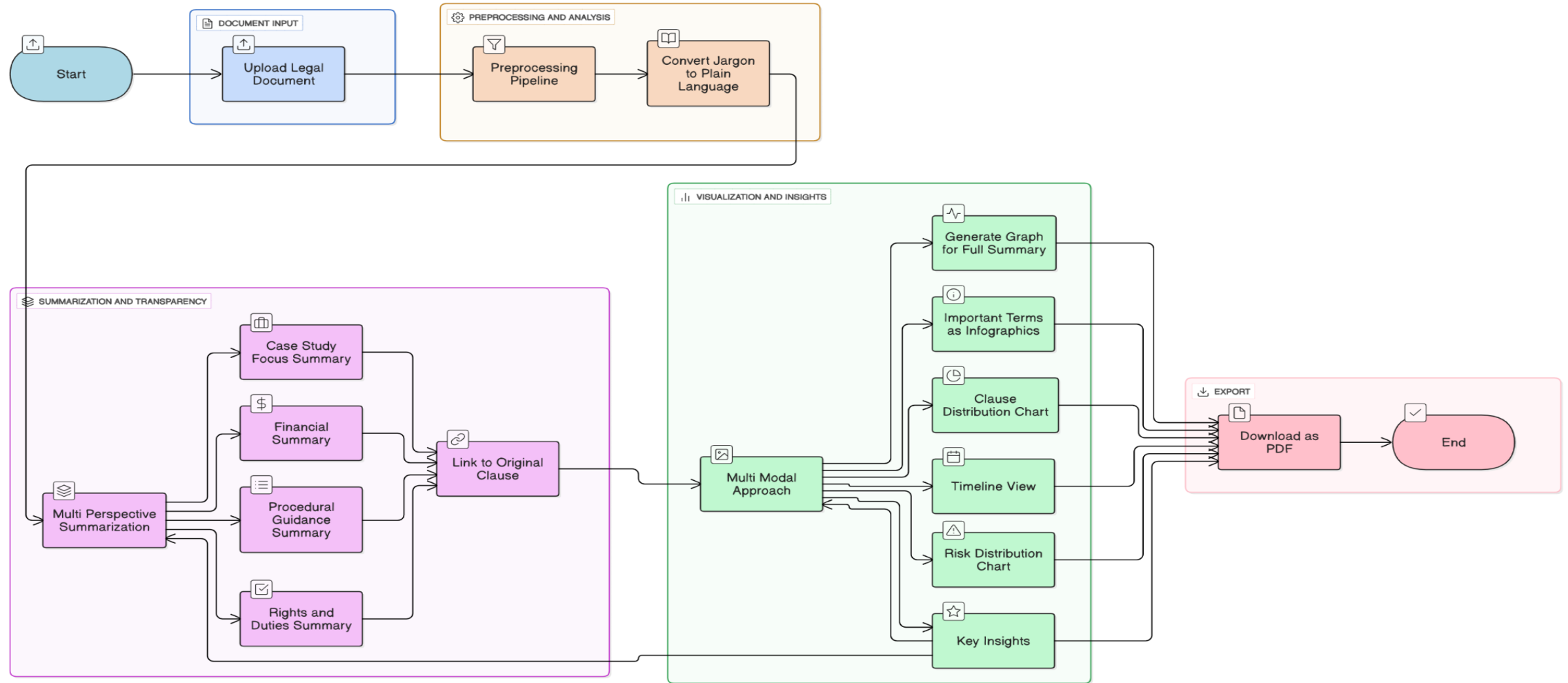
- Usability testing with non-experts
- Expert validation with legal professionals
- Comprehension improvement (pre/post-test assessments)

Data Availability

- Legal documents collected from **lawyers & law colleges** in Sri Lanka
- Covers **property law documents** (deeds, agreements, contracts, judgments)



Visual-Integrated Multi Perspective Legal Summarizer



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INTERPRETABLE LEGAL REASONING MODULE

Information Technology Specialized in Data Science

Interpretable Legal Reasoning Module

Research Gap

- Lack of Explainability in Legal AI
- Global Focus, Limited Local Context
- Absence of Structured Legal Reasoning
- Lack of Localized Datasets

Solution

- Explainable Legal Reasoning
- Localization for Sri Lanka
- Structured Argumentation Graphs
- Creation of Local Legal Dataset

Interpretable Legal Reasoning Module

Existing Systems

System	Explainability	Structured Reasoning	Localized Datasets	End-to-End System
Split_Up (Stranieri & Zeleznikow, 1995/1998)	✓	✓	✗	✗
SHYSTER / SHYSTER-MYCIN (Popple, 1993; O'Callaghan, 2003)	✓	✗	✗	✗
HYPO (Ashley, 1990)	✗	✗	✗	✗
CATO (Aleven, 1997)	✗	✗	✗	✗
PROLEG (Bench-Capon & Coenen, 1992)	✓	✗	✗	✗
Interpretable Legal Reasoning Module (Sri Lanka Property Law)	✓	✓	✓	✓

Interpretable Legal Reasoning Module

Research problem

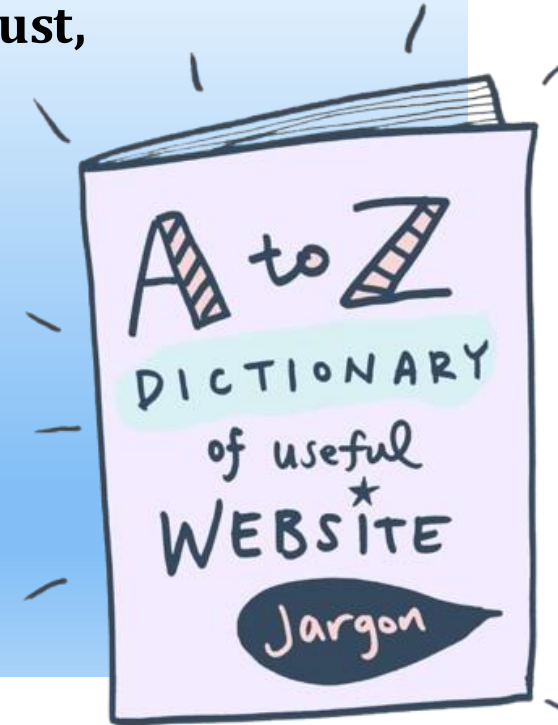
How can explainable AI techniques be integrated to provide transparent, traceable, and statute-aware reasoning for Sri Lankan property law?

Specific Objectives

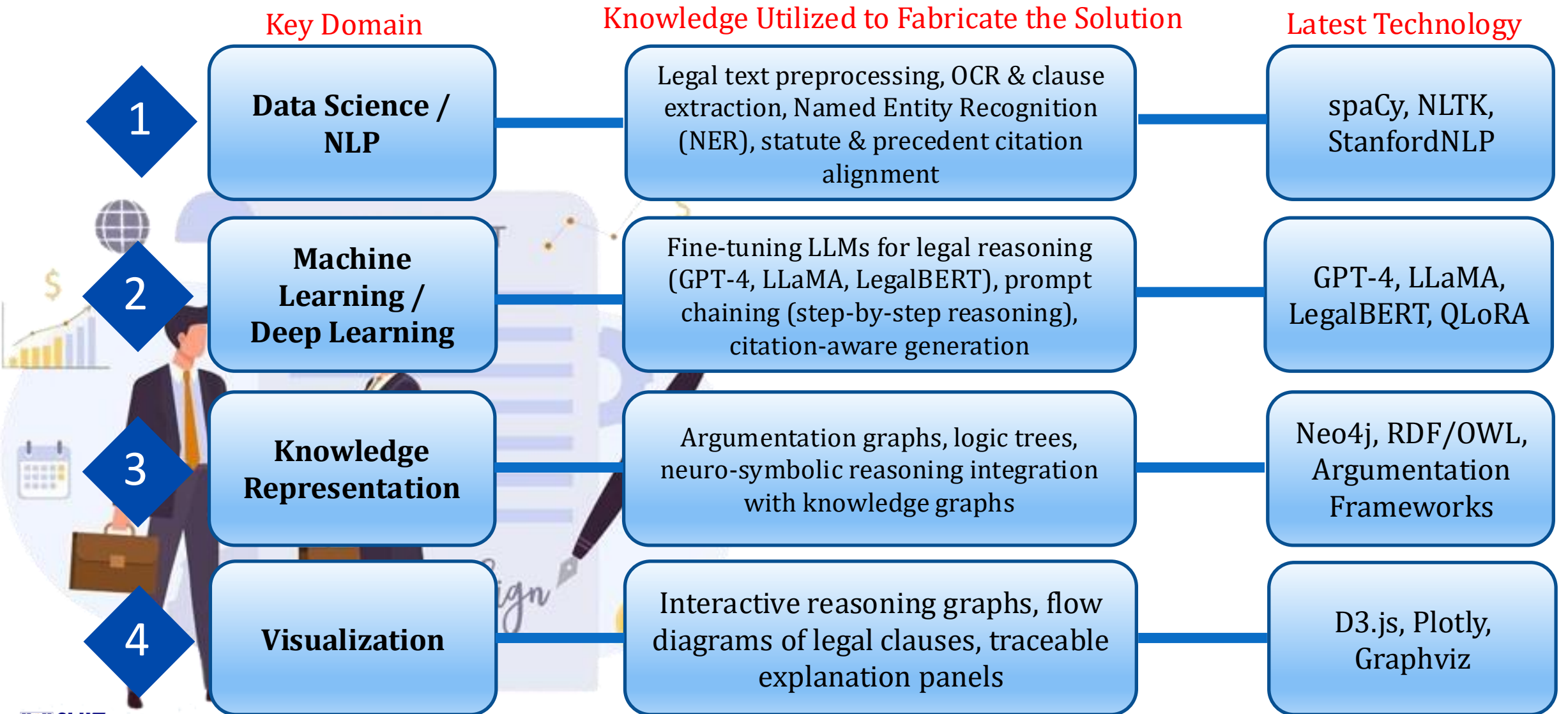
- Design and develop an explainable legal reasoning system that improves **trust, interpretability, and accessibility** of AI-driven property law analysis.

Sub Objectives

- Localize LLMs → fine-tune on Sri Lankan statutes, deeds, and case law.
- Generate **step-by-step reasoning chains** linked to statutes & precedents.
- Ensure **traceability** by citing original laws, judgments, and clauses.
- Provide **visual reasoning graphs** (logic trees, argumentation diagrams).
- Validate reasoning with **legal experts** to ensure accuracy & usability.
- Deliver a **user-friendly interface** for both professionals and non-experts.



Interpretable Legal Reasoning Module



Interpretable Legal Reasoning Module

Evaluation

Quantitative:

- Citation Accuracy (%)
- Precision & Recall
- Reasoning Chain Completeness
- Response Time (seconds)
- Usability Metrics

Qualitative:

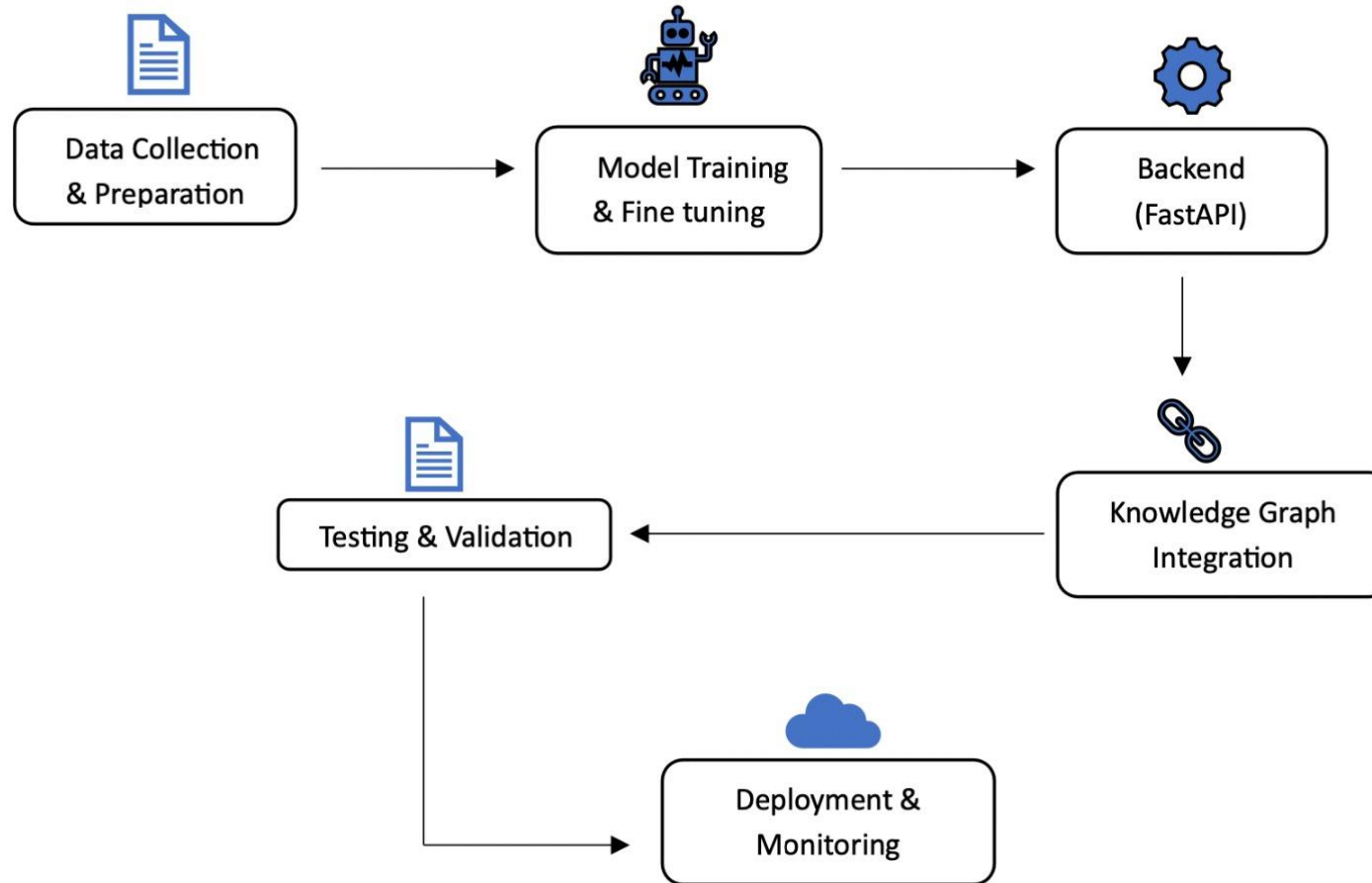
- Expert Validation
- Clarity & Transparency
- Trust & Adoption Potential
- User Experience
- Ethical & Legal Compliance

Data Availability

- Legal documents collected from **lawyers & law colleges** in Sri Lanka
- Covers **property law documents** (deeds, agreements, contracts, judgments)



Interpretable Legal Reasoning Module



REFERENCES

- [1] A. Stranieri and J. Zeleznikow, "A hybrid rule–neural approach for the automation of legal reasoning in the discretionary domain of family law in Australia," *Artif. Intell. Law*, vol. 6, no. 2–4, pp. 153–183, 1998.
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BIAS AND RISK EVALUATOR

Information Technology Specialized in Data Science

Bias & Risk Evaluator

Research Gap

- Domain specificity (property/land)
- Summary trust metric
- Explainability & human-in-the-loop evaluation

Solution

- Clause-level bias detection
- Severity labels
- A calibrated trust_score
- Highlight & visualize

Existing Systems

- CLAUDETTE → Online Terms of Service & similar consumer contracts.
- BERT & LEGAL-BERT → Transformer-based language models.

Bias & Risk Evaluator

Research problem

How can we automatically detect clause-level bias and associated risk severity in property agreements, and how can these outputs be combined into a well-calibrated, interpretable trust_score that helps end users decide whether to sign or to negotiate revisions?

Objective

Implement a Bias & Risk Evaluator for property agreements that detects clause-level bias, quantifies risk severity, and computes a calibrated trust_score to support user decision-making prior to signing.

Key domains of knowledge

- Natural Language Processing (clause segmentation, tokenization)
- Legal domain expertise (law patterns, clause semantics)
- Machine Learning & Model Calibration (transformer fine-tuning)

Bias & Risk Evaluator

Latest technologies utilized

- Pretrained transformer models (LEGAL-BERT / RoBERTa via Hugging Face)
- Hugging Face + PyTorch for fine-tuning and inference.

Evaluation

- Clause-level precision/recall/F1 per label.
- Correlate trust_score with expert rankings (Spearman's $\rho \geq 0.75$).
- Verify results with an expert.

Data Availability

- Legal documents collected from **lawyers & law colleges** in Sri Lanka
- Generate clauses with bias & risk factors using collected documents.



Bias & Risk Evaluator



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LEGAL KNOWLEDGE GRAPH BUILDER

Information Technology Specialized in Data Science

Legal Knowledge Graph Builder

Research Gap

- Fragmented land-legal data across different platforms .
- PDF-first publishing; low machine-readability.
- No common identifiers & interoperable metadata.
- Missing Sri Lanka-specific property-law ontology.
- Limited linkage of information.
- No dynamic, update-aware Sri Lanka property-law KG

Solution

- Legal Data Pipeline
- Entity and Relationship Extraction
- Knowledge Graph Construction
- Dynamic Updates and Continuous Integration
- Integrate APIs for reasoning and analytics

Legal Knowledge Graph Builder

Existing Systems

System	Integrated Data	Machine-Readable	Ontology Support	Dynamic KG Access
EUR-Lex / ELI KG	X	✓	✓	✓
Wolters Kluwer – German Court KG (ESWC'20)	X	✓	✓	X
Austria Linked Legal Data KG (AI & Law'21)	✓	✓	✓	X
Chinese Legal KG (MDPI'24, JKEM / CLKG)	✓	✓	X	X
Vietnamese Legal Cases KG (arXiv'23)	X	✓	X	X
Japanese Admin-Court KG (arXiv'25)	X	✓	✓	X
LegalVision LKG (Sri Lanka)	✓	✓	✓	✓

Legal Knowledge Graph Builder

Research problem

Sri Lanka lacks a dynamic, machine-readable legal knowledge graph for property law, with data fragmented in static, non-standardized formats

Specific Objectives

- To **design and implement a dynamic, domain-specific legal knowledge graph (KG) for Sri Lankan property law** that integrates statutes, gazettes, case law, cadastral data, and registry records into a machine-readable, continuously updated, and queryable platform.

Sub Objectives

- Build a comprehensive legal data pipeline
- Design and populate the knowledge graph
- Enable dynamic updates and continuous integration
- Integrate APIs for reasoning and analytics



Legal Knowledge Graph Builder

Key Domain

Knowledge Utilized to Fabricate the Solution

Latest Technology

1

**Data Science /
NLP**

OCR, text extraction, entity tagging,
cleaning/normalization of legal
texts, named entity recognition
(NER), relation extraction

Python (spaCy,
LegalBERT), OCR
tools

2

**Knowledge
Graph
Engineering**

Graph construction

Neo4j , MongoDB
for metadata

3

**Backend
Engineering**

API development, query access,
reasoning integration, monitoring &
logging

FastAPI(RESTAPI,
GraphQL)

Legal Knowledge Graph Builder

Evaluation

Quantitative:

- Entity Extraction Accuracy
- Relationship Mapping Accuracy
- Query Performance
- Coverage & Completeness
- Update Latency
- System Load Handling

Qualitative:

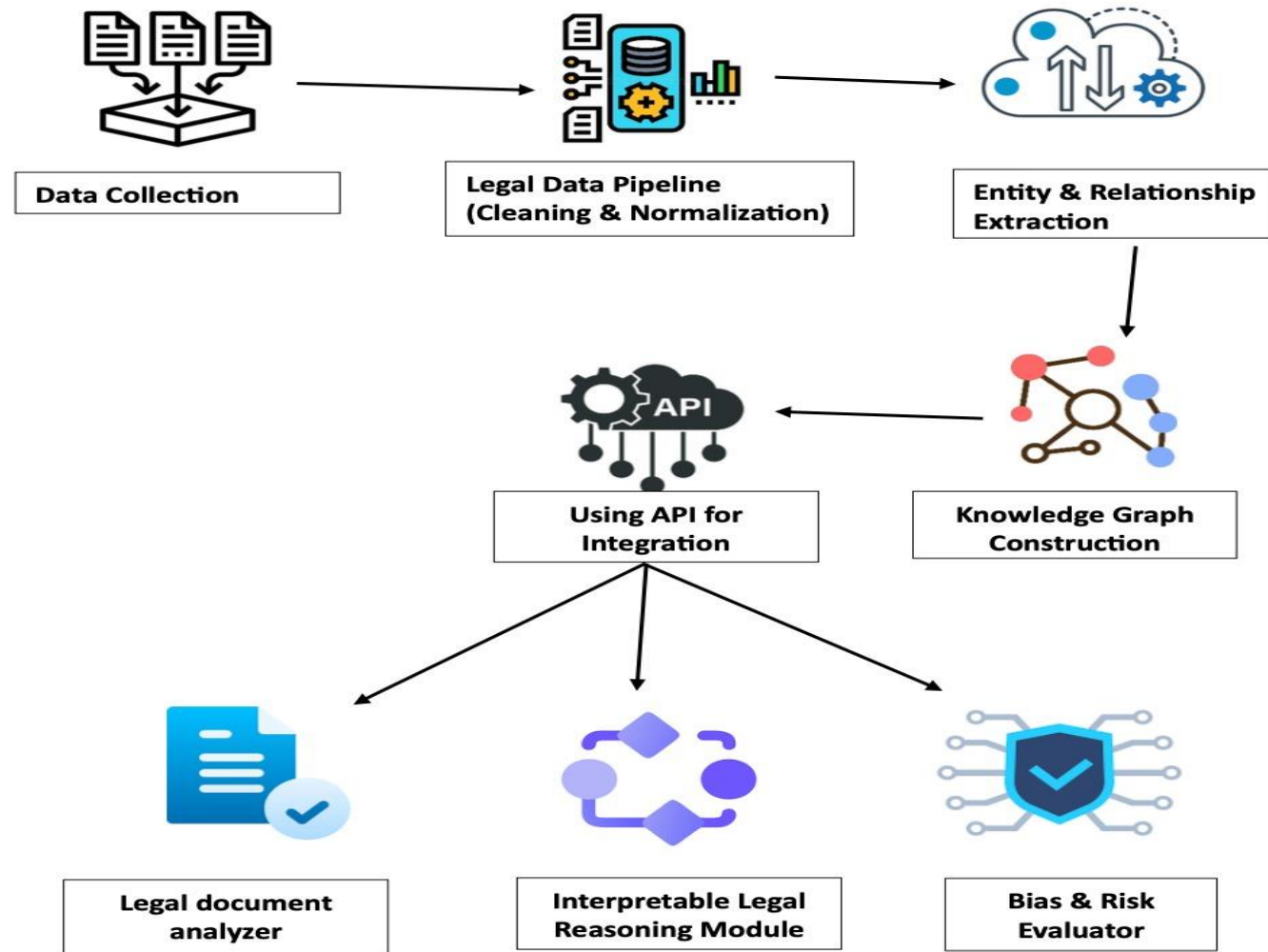
- Expert Validation
- Explainability & Transparency
- User Experience
- Trust & Adoption
- Compliance & Ethics

Data Availability

- Legal documents collected from **lawyers & law colleges** in Sri Lanka
- Covers **property law documents** (deeds, agreements, contracts, judgments)



Legal Knowledge Graph Builder



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A Knowledge-Driven AI Framework for Legal Understanding and Trust

Investment / Budget

- **Total:** 139,000 LKR
 - Travel -10,000, Data Collection - 5,000, Publication - 100,000, Internet - 24,000

Target Market & Customers

- Property Owners & Buyers
- General Citizens
- Legal professionals



Value Proposition

- **Comprehensive Summaries:** Multi-perspective (case, financial, procedural, rights & duties)
- **Risk Awareness:** Highlights risks and biases
- **Trust & Transparency:** Traceable to original clauses
- **Visuals & Accessibility:** Infographics, graphs, timelines



Revenue & Investment Recovery

- **Premium subscriptions** for advanced features (in-depth visualizations, risk analysis, and downloadable reports)
- **Free community version** via universities, legal aid centers and NGOs
- Adoption via public campaigns & institutional partnerships

A Knowledge-Driven AI Framework for Legal Understanding and Trust

